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An Explainable Machine Learning-Based Phishing Website Detection using Gradient Boosting

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***Abstract*—As the need for achieving optimal performance in prediction models, several recent and complex models have been developed. However, many of these models operate as black boxes, providing little insight into their predictive results. In recent years, three gradient-boosting methods based on Decision Trees have been recommended, i.e., XGBoost, CatBoost, and LightGBM. These methods have proven to deliver competitive performance with fast training times. Nevertheless, in critical domains like security, there is a growing need for increased transparency among stakeholders. One pressing concern in the security field is the proliferation of phishing websites. To address this issue, we propose an explainable machine learning-based approach using gradient-boosting methods with hyperparameter optimization on three phishing website datasets. Our best methods surpass the state-of-the-art phishing website detection methods, achieving accuracy rates of 97.45%, 99.16%, and 97.85% for the UCI (2015), Mendeley (2018), and Mendeley (2020) datasets, respectively. Subsequently, we implement post-hoc explainability using SHAP and LIME for the selected dataset. The experimental results indicate that three features, i.e., *length_url*, *directory_length*, and *time_domain_activation*, are consistently identified as the most influential features in the dataset. Moreover, our proposed approach demonstrates promising results for detecting phishing websites with both high accuracy and explainability.**

***Keywords*—Black-box models, explainable machine learning, gradient-boosting methods, hyperparameter optimization, phishing website detection.**

I. Introduction

The rapid growth of internet usage has led to an increase in online threats, including phishing attacks. These attacks involve the use of fake websites to deceive users into divulging sensitive information such as login credentials, personal identification numbers, and more. To mitigate the harm caused by these attacks, a machine learning-based classification approach is necessary to identify and distinguish between legitimate and phishing websites based on their characteristic features [1].

Phishing website detection has been extensively explored in numerous research studies, which have utilized a range of approaches. In particular, several classifiers have been implemented to identify phishing websites. K-Nearest Neighbors (KNN) combined with SMOTE was used to classify phishing websites with imbalanced data [2]. Feature selection was additionally applied to select the most influential features. The results indicated that the full feature set achieved an accuracy of 97.47%, while the reduced feature set still maintained a high accuracy of 94.87%.

Comparative studies in phishing website detection were conducted to evaluate the effectiveness of the classifier performance using multiple machine learning models, including SVM, Random Forest, Decision Tree, ANN, and Functional Tree meta-learners [3], [4]. The evaluation results demonstrated that the proposed methods could enhance prediction accuracy while reducing the number of features. Despite its widespread adoption, the dataset used has become outdated for developing modern phishing detection algorithms.

A URL-based approach was employed for phishing website detection, utilizing a hybrid method that combines Artificial Neural Networks (ANN) with AdaBoost models [5]. The results showed that the combined approach achieves an accuracy of 98.60%, outperforming the standalone ANN model by 0.14%. Nevertheless, the integration of a hybrid model failed to produce a notable enhancement in performance.

A URL detection technique was additionally conducted using the Long Short Term Memory (LSTM) and Recurrent Neural Network (RNN) to identify phishing and legitimate websites [6]. Besides, a Convolutional Neural Network (CNN) with a self-attention mechanism was applied for a phishing website detector with imbalanced data [7]. The dataset was balanced by employing a Generative Adversarial Network (GAN). The results reported an improved accuracy, although the proposed phishing detector lacks supplementary features for detection, such as images and HTML content.

Ensemble machine learning-based methods and deep learning-based methods were studied for phishing detection [8], [9]. The study revealed that ensemble machine learning methods consistently outperform other approaches in phishing classification compared to deep learning-based methods. The study also found that deep learning approaches have been frequently hindered by several significant challenges, including the requirement for manual architecture design, laborious parameter tuning, complexity in training, and inadequate accuracy.

In addition, a major drawback of deep learning-based models is that they are often regarded as black-box models, lacking the ability to provide insightful explanations for their predictions. While they do offer superior prediction accuracy, their increasing complexity compromises model transparency [10]. Therefore, this paper aims to propose explainable machine learning-based phishing website detection using gradient-boosting algorithms due to their relatively fast training times and minimal parameter tuning requirements [11]. Additionally, post-hoc explainability methods, including SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanation), are applied to elucidate how the model generates predictions.

This paper outlines the main objective as follows: to evaluate the highest accuracy achievable using gradient-boosting algorithms for detecting phishing websites and for comparing to the state-of-the-art methods, to integrate post-hoc explainability methods into the best-performing model, and to assess the explanatory capabilities of the model after applying post-hoc explainability, thereby enhancing transparency in phishing website detection.

II. Materials and Methods

A. Phishing Website Dataset

This study utilizes publicly available phishing website datasets retrieved from the UCI (2015) machine learning repository [12], Mendeley (2018) [13], and Mendeley (2020) [14]. These datasets comprise several features that are categorized into binary classes, namely phishing or legitimate. A comparison of these three common phishing datasets is presented in Table I, and a more detailed description of each dataset can be found in [8].

B. Synthetic Minority Oversampling Technique (SMOTE)

SMOTE operates by employing an oversampling approach to generate new synthetic samples for the minority class, thereby rebalancing the class distribution. This involves creating new synthetic data through interpolation between selected minority class samples based on their k-nearest neighbors. The amount of synthetic data generated is determined by the required percentage of oversampling, which is calculated based on the existing class distribution. New samples are then generated through interpolation using the following formula [15]

$$S = S_i + (S_k - S_i) * \alpha \tag{1}$$

where S represents the new synthetic sample, S_i denotes the original sample, S_k denotes the nearest neighbor sample, and α is a random number within the range $[0, 1]$. Finally, the newly generated samples are incorporated into the minority class.

C. Gradient Boosting Algorithms

1) eXtreme Gradient Boosting (XGBoost): Gradient boosting is an ensemble-based method that combines multiple weak classifiers or regressors, such as decision trees, to produce a stronger predictive model [11]. XGBoost, in particular, constructs an additive extension of the objective function by iteratively minimizing the loss function. Notably, since XGBoost exclusively relies on decision trees as its base classifiers, various loss function variations are employed to regulate the complexity of the trees [16].

XGBoost includes an additional regularization hyperparameter, known as shrinkage, which reduces the step size in the additive extension. Furthermore, the complexity of the trees can be controlled through other strategies, such as limiting the tree depth. Additionally, XGBoost employs randomization techniques to mitigate overfitting and accelerate training speed.

2) Light Gradient Boosting Machine (LightGBM): The development of LightGBM was motivated by the limitations of XGBoost, particularly its scalability issues when handling large datasets with numerous samples and features. To address these limitations, LightGBM introduces several notable enhancements, including leaf-wise tree growth, a histogram-based splitting algorithm, and Gradient-based One-Side Sampling (GOSS) for efficient sample weighting [11].

GOSS is a subsampling technique used to create training sets for building base trees in an ensemble. Additionally, LightGBM offers L1 and L2 regularization, as well as numerous hyperparameters that can substantially impact the algorithm's effectiveness [17].

3) Categorical Boosting (CatBoost): CatBoost is designed to efficiently process categorical features during training and mitigate the prediction shift issue that affects most gradient boosting implementations, potentially compromising model performance [11]. This method introduces two innovative concepts: ordered target statistics and ordered boosting, which utilize permutations. Unlike XGBoost and LightGBM, CatBoost employs symmetric trees as its basis, where each level is balanced using identical splitting conditions.

TABLE I
COMPARISON OF UCI (2015), MENDELEY (2018), AND MENDELEY (2020) PHISHING WEBSITE DATASETS [8]

Dataset	Number of Samples	Phishing Samples	Legitimate Samples	Number of Features	Type of Features
UCI (2015)	11055	4898	6157	30	Categorical
Mendeley (2018)	10000	5000	5000	48	Mixed
Mendeley (2020)	88647	30647	58000	111	Mixed

Notably, CatBoost automatically selects the optimal learning rate during training, eliminating the need to consider it as a hyperparameter. However, unlike its counterparts, CatBoost has limitations in handling categorical data and missing values, which are treated differently compared to XGBoost and LightGBM. Specifically, CatBoost does not support L1 regularization, which can negatively impact the model's generalization ability [16].

D. Hyperparameter Optimization

1) Randomized Search (RS): Rather than exploring all possible combinations of hyperparameter values, RS samples a fixed number from a specific distribution [18]. This approach is generally faster than grid search, with a computational complexity of $O(n)$ [19]. Each sampled combination represents a unique set of hyperparameters. The process then involves training and evaluating the model until the most optimal set of hyperparameters is found.

2) Bayesian Optimization (BO): BO leverages Gaussian processes to dynamically update previous values by incorporating knowledge from past parameters. This approach yields fewer and faster iterations [19]. By constructing a probability model of the objective function, BO selects hyperparameters for evaluation in the actual objective function. The Bayes formula is then used to compute the posterior probability distribution. The process continues with obtaining the mean and variance of accuracy for each hyperparameter at each point of value.

3) Tree-based Parzen Estimator (TPE): TPE is a variant of traditional Bayesian Optimization, which transforms the configuration space into a non-parametric density distribution [18]. The configuration space can be represented using various distributions, including uniform, discrete uniform, or logarithmic uniform. TPE's primary goal is to identify the optimal combination of hyperparameters for machine learning models that achieve the best performance in specific tasks [19].

E. Explainable Machine Learning

1) SHapley Additive exPlanations (SHAP): SHAP applies the concept of Shapley values from game theory to assign contributions to each feature concerning the model's output. By calculating SHAP values, it becomes clear how each feature contributes to the model's prediction, enabling explanations of model behavior. Moreover, since SHAP values are model-agnostic, they can be used to explain the behavior of various machine learning models, including linear models, neural networks, and decision trees [20].

2) Local Interpretable Model-Agnostic Explanation (LIME): LIME is a technique designed to provide explanations for individual predictions made by complex, opaque machine learning models. At its core, LIME involves training simple and interpretable surrogate models on slightly modified samples that are similar to the sample of interest. By analyzing how these surrogate models behave around this specific sample, LIME seeks to approximate the behavior of the original complex model in the local vicinity of the prediction [21].

F. Conceptual Framework of the Proposed Method

The conceptual framework of the proposed explainable machine learning model for phishing website detection is illustrated in Fig. 1. We use three common public datasets, i.e., UCI (2015), Mendeley (2018), and Mendeley (2020). As the datasets are imbalanced, particularly UCI (2015) and Mendeley (2020), SMOTE is applied to balance the class. Additionally, to address issues with missing values and duplicated rows in the datasets, data preprocessing is performed before applying the classifiers. Initially, three machine learning models, i.e., XGBoost, LightGBM, and CatBoost, are implemented and trained on preprocessed features. Furthermore, we propose hyperparameter optimization methods, including RS, TPE, and BO, to enhance the performance of gradient-boosting algorithms. The proposed method's performance is then evaluated using accuracy metrics.

Once the best-performing model is selected, then post-hoc explainability is applied to generate explanations. For this purpose, the dataset Mendeley (2020) is selected due to its large number of features and samples. To gain insights into the critical factors driving the model's overall predictions, we employ global explanation using SHAP values. Additionally, local explanation using LIME is utilized to provide sample-level explanations for individual model predictions. These techniques are chosen due to their ability to provide robust

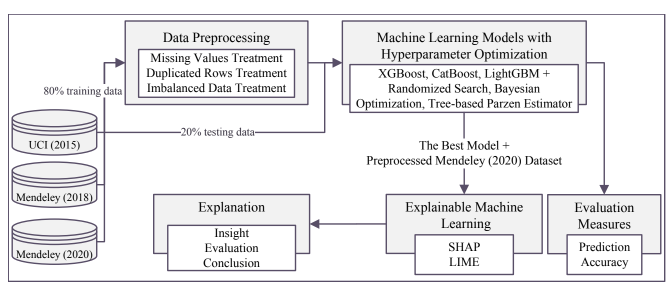

Fig. 1. Conceptual framework of the proposed explainable machine learning model for phishing website detection.

and interpretable explanations of complex models.

III. Result and Discussion

A. Experimental Setup

The experiments were carried out in a Google Colaboratory environment using Python 3.10.6. The dataset was divided into 80% for training and 20% for testing based on the Pareto principle, with 10-fold cross-validation employed to ensure robust results [22]. To determine the optimal hyperparameter for each algorithm, the specified hyperparameter values for tuning are given in Table II, Table III, and Table IV.

B. Performance Results

Our experiments demonstrate that 21 out of 27 (77.78%) models exhibit improved performance when making predictions on data after undergoing hyperparameter tuning using RS, TPE, and BO (see Table V). These findings are consistent with previous research indicating that XGBoost, LightGBM, and CatBoost significantly benefit from hyperparameter tuning [23]. Notably, we have observed the largest performance decrease in XGBoost's accuracy on the UCI (2015) dataset when using BO, resulting in a 1.18% decrease. In contrast, the highest performance increase has been achieved with LightGBM's accuracy on the Mendeley (2020) dataset using TPE, yielding a 1.16% improvement.

Following hyperparameter tuning, CatBoost consistently achieves the highest accuracy across all datasets. Specifically, it outperforms other models on the UCI (2015) dataset when tuned with RS, on the Mendeley (2018) dataset with TPE, and on the Mendeley (2020) dataset with RS. These results align with prior research suggesting that CatBoost has been considered the most accurate implementation compared to XGBoost, LightGBM, and SnapBoost, particularly excelling with categorical data [23]. Overall, our findings confirm that CatBoost outperforms all gradient-boosting algorithms.

Lastly, our best accuracy results using CatBoost (see Table V in bold) are compared to state-of-the-art methods on the same phishing website dataset. A comprehensive comparison is shown in Table VI, revealing that our results consistently outperform other studies in all datasets.

TABLE II
HYPERPARAMETER TUNING VALUES FOR XGBOOST

Hyperparameter	Initial Value	Tuning Value
learning_rate	0.3	log-uniform(0.01, 0.3)
max_depth	6	[2, 3, 5, 7, 10, 100]
gamma	0	uniform(0, 3)
colsample_bylevel	1	[0.15, 0.5, 0.75, 1.0]
subsample	1	[0.15, 0.5, 0.75, 1.0]
alpha	0	uniform(0, 1)
lambda	1	uniform(0, 3)

TABLE III
HYPERPARAMETER TUNING VALUES FOR LIGHTGBM

Hyperparameter	Initial Value	Tuning Value
learning_rate	0.1	log-uniform(0.01, 0.3)
num_leaves	31	[3, 7, 15, 31, 127, 1024]
top_rate	0.2	uniform(0.1, 0.5)
other_rate	0.1	uniform(0.05, 0.2)
feature_fraction_bynode	1	[0.15, 0.5, 0.75, 1.0]
reg_alpha	0	uniform(0, 1)
reg_lambda	0	uniform(0, 3)

TABLE IV
HYPERPARAMETER TUNING VALUES FOR CATBOOST

Hyperparameter	Initial Value	Tuning Value
learning_rate	0.03	log-uniform(0.025, 0.3)
max_depth	6	[2, 3, 4, 5, 8, 10]
leaf_estimation_iterations	10	[1, 10]
L2_leaf_reg	3	uniform(0, 10)
colsample_bylevel	1	[0.2, 0.4, 0.6, 0.8, 1.0]

C. Explanation Results

For the implementation of post-hoc explainability and evaluation, we selected the Mendeley (2020) dataset due to its quantity of features and samples, as shown in Table I [8]. Additionally, we selected CatBoost as the model for this task, given its superior accuracy performance.

Fig. 2 presents a compilation of summary plots using SHAP values to determine the average contribution of feature values across all possible combinations of features. The features are prioritized according to their importance, with a heat map-like visualization used to illustrate their relative weights. In this representation, red indicates high-importance features, blue signifies low-importance features, and purple marks those with moderate importance. The greater the disparity in

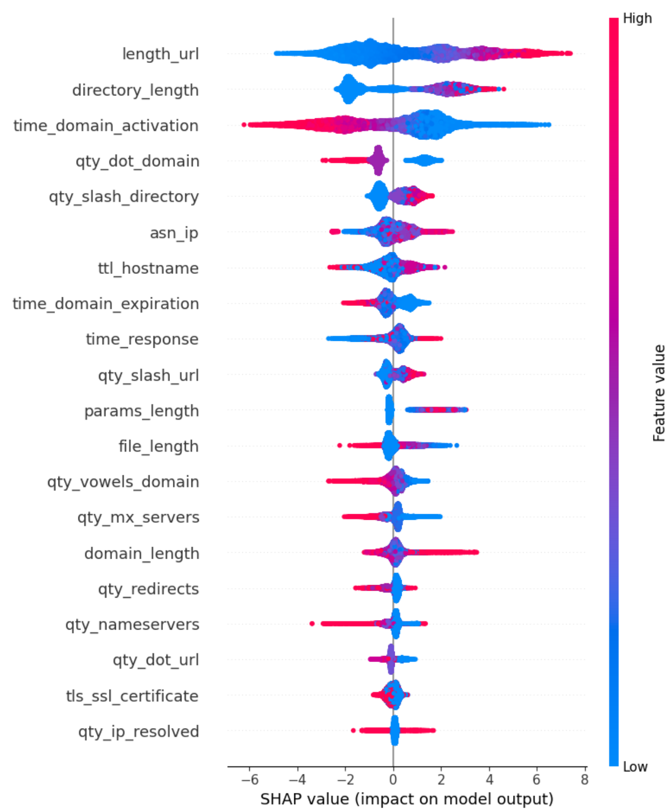

Fig. 2. SHAP explanation results.

TABLE V
ACCURACY PERFORMANCE EVALUATION WITH THE PERCENTAGE OF GAIN/LOSS (IN BRACKET) COMPARED TO THE BASELINE

Dataset	Model	Baseline (No Tuning)	Tuned with Hyperparameter Optimization		
			RS	TPE	BO
UCI (2015)	XGBoost	97.29%	97.00% (-0.29%)	96.96% (-0.33%)	96.11% **(-1.18%)**
	CatBoost	97.26%	**97.45%** (+0.19%)	97.37% (+0.11%)	97.19% (-0.07%)
	LightGBM	96.87%	97.42% (+0.55%)	96.56% (-0.31%)	96.06% (-0.81%)
Mendeley (2018)	XGBoost	98.08%	98.12% (+0.04%)	98.90% (+0.82%)	99.00% (+0.92%)
	CatBoost	98.20%	98.26% (+0.07%)	**99.16%** (+0.96%)	99.11% (+0.91%)
	LightGBM	98.02%	98.25% (+0.23%)	98.95% (+0.93%)	98.80% (+0.78%)
Mendeley (2020)	XGBoost	97.07%	97.84% (+0.77%)	97.83% (+0.76%)	97.34% (+0.27%)
	CatBoost	97.09%	**97.85%** (+0.76%)	97.83% (+0.26%)	97.23% (+0.13%)
	LightGBM	96.68%	97.47% (+0.76%)	97.84% **(+1.16%)**	97.79% (+1.11%)

color, the more consistent the feature is with predicting the label. The horizontal axis represents the SHAP values, where positive values indicate that a feature will result in a higher predicted value (i.e., phishing), and vice versa.

The most important feature is *length_url*, suggesting that as the URL length of a website increases, it tends to result in a prediction that the website is phishing, and conversely if the URL length is low, it tends to result in a prediction that the website is legitimate. The summary plot graph reveals that *length_url*, *directory_length*, *time_domain_activation*, *qty_dot_domain*, and *qty_slash_directory* are the top-5 features with the highest level of importance for this dataset.

Fig. 3 shows the feature importance with predictions indicating legitimate websites that align with ground truth. We have observed that *length_url* has the highest importance with a value of less than 18 characters, followed by *directory_length* with a value of 0 (indicating no directory in that sample). Additionally, the sample lacks equal signs, asterisks, commas, and exclamation marks in the file section. Furthermore, the URL format is not in the form of an IP address. Therefore, the model confidently classifies it as a legitimate website, which aligns with the sample's ground truth.

Fig. 4 displays the feature importance for predictions indicating phishing websites. We observe that *length_url* has the highest importance, with a value exceeding 49 characters, followed by the absence of spaces, exclamation marks, and dollar signs in both the directory and the entire URL. The sample has a directory length of more than 24 characters. Consequently, the model confidently classifies it as a phishing website, which aligns with the sample's ground truth.

TABLE VI
PERFORMANCE COMPARISON OF OUR METHOD WITH OTHER STUDIES

Dataset	Study	Accuracy
UCI (2015)	Our result	**97.45%**
	[3]	97.30%
	[4]	97.19%
	[9]	97.16%
Mendeley (2018)	Our result	**99.16%**
	[4]	98.51%
	[5]	98.60%
	[9]	98.58%
Mendeley (2020)	Our result	**97.85%**
	[8]	96.94%
	[9]	97.39%

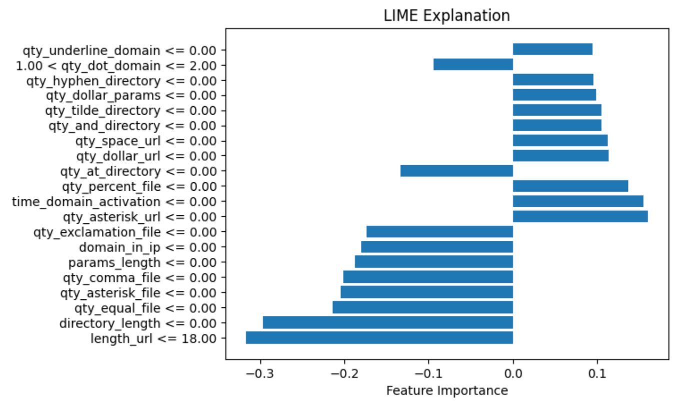

Fig. 3. LIME legitimate sample explanation results.

Based on LIME interpretation, we can conclude that *length_url* and *directory_length* play a significant role in determining whether a website is phishing or not. For both samples representing predictions consistent with the ground truth, it is evident that predictions aligned with the tendencies of *length_url* and *directory_length* values result in predictions that match the ground truth. The primary determinants for predicting phishing websites in this dataset are *length_url* and *directory_length*, with *time_domain_activation* serving as a complementary feature to enhance phishing websites detection [1].

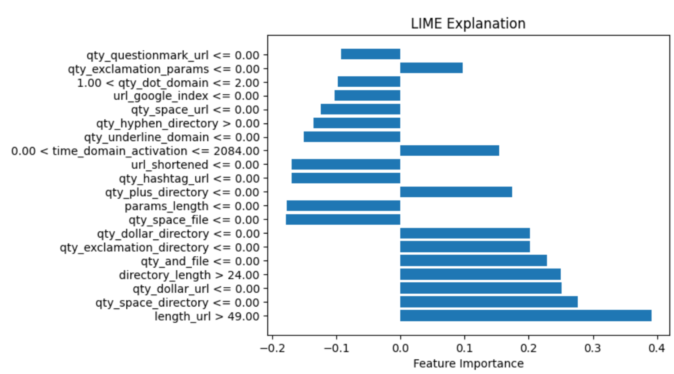

Fig. 4. LIME phishing sample explanation results.

D. Performance Evaluation of Reduced Features

Furthermore, we evaluate the model using only the three most important features, namely *length_url*, *directory_length*, and *time_domain_activation* with the same hyperparameter tuning values as specified in Table IV. The results show an accuracy of 94.34%. Notably, despite reducing the number of features by 97.29%, the accuracy only decreases by 3.58% compared to using the full set of features. This suggests that these three features are sufficient and suitable for consideration in predicting phishing websites. Due to the small number of features selected, the proposed method is considered highly efficient.

IV. Conclusion

This study demonstrates the effectiveness of explainable machine learning-based phishing website detection using gradient-boosting algorithms by overcoming black-box models' limitations and providing high explainability. We utilize three publicly available datasets from UCI (2015), Mendeley (2018), and Mendeley (2020) to classify websites as either phishing or legitimate. The choice of gradient-boosting algorithms is attributed to their fast training times and minimal parameter tuning requirements, making them well-suited for this task. Our experimental results show that our proposed methods surpass the state-of-the-art methods in phishing website detection, achieving accuracy rates of 97.45%, 99.16%, and 97.85% for the UCI (2015), Mendeley (2018), and Mendeley (2020) datasets, respectively. Following the selection of the best-performing model, we apply post-hoc explainability techniques using SHAP and LIME to enhance transparency from the black-box models. The results indicate that three key features, i.e., *length_url*, *directory_length*, and *time_domain_activation*, are the primary drivers in predicting phishing websites. Our proposed approach demonstrates promising results for detecting phishing websites with both high accuracy and explainability, thereby improving the transparency of machine learning models. Future research remains possible using other ensemble learning models and post-hoc explainers involving Partial Dependence Plot and Anchors.

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